Optimal Execution of Large NSE Equity Orders using Deep Reinforcement Learning

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Code, trained agents & results: `https://github.com/Harsh-nandanshukla/rl-optimal-execution`

Abstract. A fund that must sell a large block of shares within one trading day faces a trade-off: selling fast moves the price against it (market impact), selling slowly exposes it to price risk. We cast this as a finite-horizon MDP and train DQN (written from scratch) and PPO agents to choose how much to sell in each of 25 fifteen-minute intervals. Because the environment's return equals the Almgren–Chriss (AC) objective exactly, we can test RL against a known optimum. On 10,000 held-out simulated days, both agents recover the AC schedule to within 0.14–0.17 bps (98% of AC's gain over TWAP). When prices carry a predictable short-term drift, the agents beat AC by 14–15 bps and a hand-written signal rule by 4–5 bps; hiding the signal removes the edge. On real NSE minute data (2023–26), the signal-free agents match AC and beat TWAP by 8 bps, while signal-trained agents do not transfer.

1 Problem and Motivation

Institutional orders are far larger than the liquidity available at any instant, so they are split into child orders over the day. Each child order pays a *temporary impact* cost that grows with trading speed, and every sale also lowers the price permanently. Trading slowly avoids impact but leaves inventory exposed to price volatility. Almgren and Chriss [1] solved this trade-off in closed form under a random-walk price; industry benchmarks such as TWAP and VWAP ignore risk entirely. Our question is whether a learned policy can (i) rediscover the analytical optimum when it exists, and (ii) beat it when prices are partly predictable, which the static AC schedule cannot exploit. RL fits because decisions are sequential and the cost is only known at the end of the day.

2 MDP Formulation

Units are normalised: arrival price S_0=1, order size X=1; costs $\times 10^4$ are basis points (bps). A day has N=25 steps of length τ=1/25. Holding x_k shares and selling n_k, the dynamics are

$$S_{k+1} = S_k + \sigma\sqrt{\tau}\,\xi_k + \mu_k\tau - \gamma n_k, \qquad \tilde{S}_k = S_k - \varepsilon - \eta\, n_k/\tau,$$

where $\tilde{S}_k$ is the execution price, $\xi_k \sim \mathcal{N}(0,1)$, σ=2%/day, η=0.002, γ=0.001, half-spread ε=0.5 bps. The drift μ_k is zero in Experiment A and an AR(1) process (ϕ=0.8, s.d. 0.04) in Experiment B.

State (4-dim, all $O(1)$): inventory left, time left, observed drift μ_k (set to 0 in the ablation), last normalised price move. **Action** (12 discrete): sell $m \cdot x_k/(N-k)$ with $m \in \{0, 0.25, \ldots, 2, 2.5, 3, 4\}$; m=1 is "TWAP pace from here". Unsold shares are force-sold at the last step. **Reward**: mark-to-market P&L over the interval minus a risk penalty $\lambda\sigma^2\tau x_k^2$, with λ=25 and discount γ_{RL}=1. By construction the episode return equals

$$\textstyle\sum_k r_k = -\big(\mathrm{IS} + \lambda\sigma^2\tau\sum_k x_k^2\big),$$

i.e. minus the AC objective (implementation shortfall plus risk). We verified this identity to 10^{-12} in code. Hence in a random walk the optimal policy *is* the AC schedule $x_k^\star = \sinh(\kappa(T-t_k))/\sinh(\kappa T)$ ($\kappa T \approx 2.25$), which gives Experiment A a known right answer.

Table 1: Held-out test results (bps, lower is better). First row: AC objective; other rows: objective minus AC on the same paths. Baselines: 95% CI over paths (real: clustered over 803 dates). Agents: mean ± t-interval over 5 seeds.

Strategy	Exp A random walk	Exp B AR(1) drift	Real NSE 2023–26
AC objective	49.34±1.69	49.25±2.38	55.23±4.31
AC on action grid	+0.26±0.01	+0.27±0.01	+0.30±0.04
TWAP	+7.60±0.61	+7.59±0.90	+8.24±1.18
VWAP	—	+7.23±0.90	+11.84±1.23
Signal rule (heuristic)	—	−9.85±0.80	—
Immediate	+451.2	+451.3	+445.3
DQN, Exp A agent	**+0.14**±0.04	—	**+0.15**±0.10
PPO, Exp A agent	**+0.17**±0.07	—	**+0.17**±0.09
DQN, signal observed	—	**−13.94**±1.16	+8.75±5.38
PPO, signal observed	—	**−15.01**±0.99	+2.22±0.90
DQN, signal hidden	—	+1.07±0.51	+2.22±0.48
PPO, signal hidden	—	+0.70±0.90	+2.58±0.65

3 Method

DQN [2] (from scratch, PyTorch). MLP 4→128→128→12, replay buffer 100k, Double-DQN targets, soft target update (τ=0.005), Huber loss, Adam (lr 5×10^{-4}), gradient clipping, ε decayed 1→0.05 over 20k steps. **PPO** [3] (Stable-Baselines3 [4], permitted by the TAs) uses the same 12 actions, a 128×128 network and γ=1; hyperparameters were fixed in advance, not tuned. **Variance-reduced training reward.** Price noise (∼100 bps/day) swamps the 7.6 bps AC–TWAP gap, so on synthetic paths we train on the conditional expectation of the noise term. This is unbiased under γ=1 and leaves the optimal policy unchanged; all evaluation uses realised costs. **Protocol.** Each configuration is trained for 300k steps (12,000 days) with 5 seeds on CPU. The best checkpoint is chosen on 1,000 fixed validation paths. Models are then committed and frozen with a SHA-256 manifest, and the held-out test set (10,000 paths) is evaluated *once*. All strategies see identical paths, so we report paired differences with 95% CIs, plus a t-interval across seeds. Baselines: Immediate, TWAP, VWAP, closed-form AC, and AC snapped to the 12-action grid ("AC on grid"). **Real data.** 1-minute NSE bars for ADANIPORTS, BAJAJ-AUTO and COALINDIA (Kaggle, 2015–26) are cut into clock-time 15-min buckets, keeping full sessions only. Split: train ≤2021, validation 2022, test 2023-01 to 2026-04 (803 dates, 2,409 stock-days). Impact is still simulated on the real price paths; VWAP uses the training-period volume profile (no look-ahead), and CIs are clustered by date because the three stocks share dates.

4 Results

Experiment A: RL recovers the optimum. In a pure random walk both agents land within 0.14 (DQN) and 0.17 bps (PPO) of AC on 10,000 test days, keeping 98.1% and 97.7% of AC's 7.6 bps advantage over TWAP (Table 1). Their mean inventory curves are visually indistinguishable from AC (Fig. 1a): holdings stay within about 1–2% of the order at every step, and the error is almost entirely a fixed bias with near-zero path-to-path dispersion. The agents therefore learned a static, AC-shaped schedule and correctly ignore price noise, which is exactly

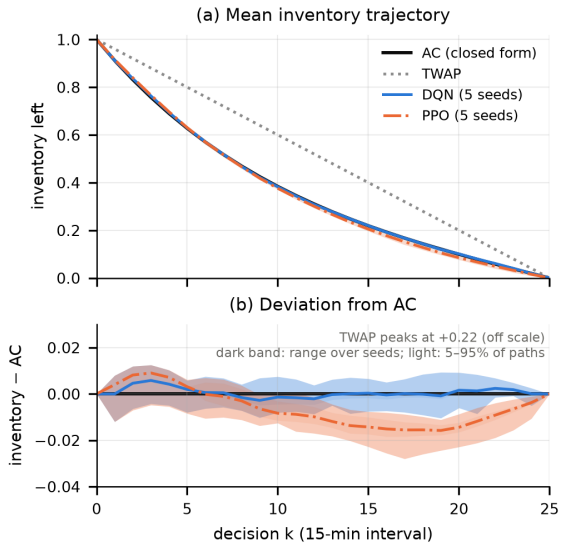


Figure 1: Exp A (test). (a) Mean inventory of DQN and PPO (5 seeds each) vs. AC and TWAP. (b) Deviation from AC; dark band: range over seeds, light: 5–95% of paths.

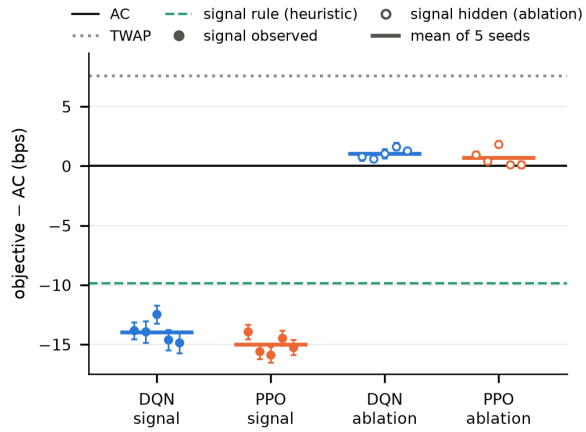


Figure 2: Exp B (test, 10,000 paths): objective minus AC per seed (dots, 95% CI) and mean (bar). Signal observed vs. hidden (ablation); TWAP and the signal rule are reference lines.

what theory prescribes here. 9 of 10 agents beat AC snapped to the action grid by tracking (+0.26 bps), so most of the remaining gap is the 12-action discretisation rather than failed learning. DQN was slightly closer to AC (trajectory RMSE 0.006 vs. 0.011) though noisier during training.

Experiment B: RL beats the optimum when it can see a signal. With a predictable AR(1) drift, agents that observe it beat AC by $13.9_{\pm 1.2}$ (DQN) and $15.0_{\pm 1.0}$ bps (PPO), i.e. about 30% of the total cost, and every per-seed CI excludes zero (Fig. 2). They also beat a hand-written rule ("sell slower when drift is positive") by 4–5 bps, so RL learns more than the obvious heuristic. Mechanistically, the agents cut mean IS from 30.7 to 10–13 bps at the price of a higher risk term (21–25 vs. 18.5), while the tail cost improves ($\text{CVaR}_{5\%}$ 242–254 vs. 281 bps). A regression of trade deviation on observed drift gives slopes of -1.70 (DQN) and -1.50 (PPO) vs. -1.41 for the rule, which confirms the agents are trading on the signal. **Ablation.** Retraining with the drift hidden from the state removes the edge entirely: the ablated agents score +1.07 and +0.70 bps relative to AC, and the matched-seed difference is -15.0 ± 1.5 (DQN) and -15.7 ± 1.5 bps (PPO). The last-price-move feature remains visible and correlates weakly (≈ 0.3) with the drift, yet yields no measurable advantage.

Real NSE data: transfer. On 803 held-out trading days (2,409 stock-days), the signal-free Exp A agents match AC

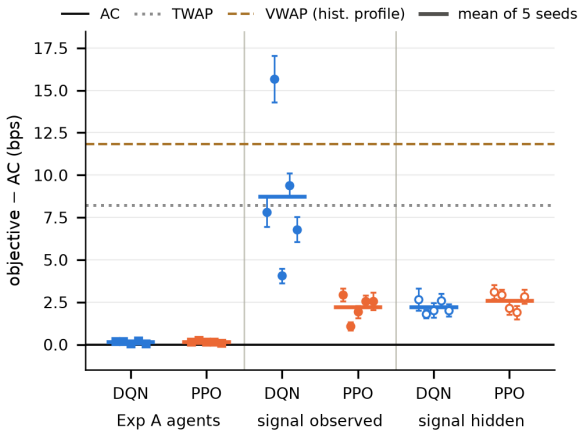


Figure 3: Real NSE test days 2023–26 (simulated impact): objective minus AC for every agent (dots, date-clustered 95% CI) and seed mean (bar).

(+0.15/+0.17 bps) and beat TWAP by 8.1 and historical VWAP by 11.7 bps, with the same sign on each stock (Fig. 3). Signal-trained agents do *not* transfer: real days carry no drift feature, so they always observe $\mu = 0$, a region training never targeted. There PPO loses 2.2 bps and DQN holds too much inventory, losing 8.8 bps on average with a wide seed spread (+4 to +16; two seeds are worse than TWAP). VWAP trails TWAP because in the AC impact model cost depends on trade size but not on market volume, so trading heavily at the open and close is only penalised (a modelling limitation).

5 Discussion and Limitations

Three results follow. (1) Given the right reward, a generic RL agent rediscovers a classical closed-form solution, which validates both the environment and the learners. (2) RL's value lies where closed forms stop, namely exploiting state information that a static schedule cannot use. (3) That value is conditional on the signal existing at deployment: a signal-dependent policy becomes fragile when the signal vanishes, so in practice the signal-free agent is the safe default. Limitations: impact is simulated with a linear AC model (no order book, no volume-dependent impact), σ is fixed, the synthetic signal is noiseless and correctly specified (an upper bound on real gains), we tested a single λ, only 5 seeds per arm, the ablation still exposes the last price move, and Exp A and Exp B test paths share their noise draws. Natural extensions are a noisy or estimated real signal, volume-dependent impact, and a λ sweep tracing the cost–risk frontier.

6 Conclusion

We built a verified execution simulator, a from-scratch DQN, and a leakage-free evaluation pipeline (paired paths, seeds, a freeze-then-test-once protocol, and date-clustered CIs on real data). RL matches the Almgren–Chriss optimum where it is optimal, beats it by $\sim$30% when a predictive signal is observable, and loses the edge exactly when the signal is removed.

References

[1] R. Almgren, N. Chriss. Optimal execution of portfolio transactions. *J. Risk*, 3(2), 2000.

[2] V. Mnih et al. Human-level control through deep RL. *Nature*, 518, 2015; H. van Hasselt et al. Deep RL with Double Q-learning. AAAI, 2016.

[3] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[4] A. Raffin et al. Stable-Baselines3: reliable RL implementations. *JMLR*, 22, 2021.